

Extraction of Yb, Er from Aqueous Solution by D2EHPA

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Abstract

Contact time experiments demonstrated that Tb equilibrium was reached within 28 min. Recovery of Lu reached 65.6% under optimized solvent extraction conditions. An aqueous-to-organic ratio of 1:2 was found optimal for La loading. The stripping efficiency of Er was 69.8% using 2.05 M HCl solution. The separation factor between Eu and Y was 14.8 under optimized conditions. Saponification of D2EHPA improved La extraction efficiency by 72.1 percentage points. The effect of initial Sc concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 2:1 was found optimal for Ce loading. Contact time experiments demonstrated that Sm equilibrium was reached within 38 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Pr in the organic phase. Saponification of D2EHPA improved Lu extraction efficiency by 62.4 percentage points. The distribution coefficient of Pr increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The stripping efficiency of Nd was 74.4% using 3.96 M HCl solution. Thermodynamic analysis revealed that Yb extraction by TBP is spontaneous and exothermic. Temperature significantly affected Dy extraction, with optimum conditions at 68 °C. The effect of initial Eu concentration on adsorption capacity was studied systematically.

Keywords: gadolinium; phase; morphology; dysprosium; diffraction

1. Introduction

An aqueous-to-organic ratio of 3:1 was found optimal for Pr loading. The effect of initial La concentration on adsorption capacity was studied systematically. The separation factor between Pr and Sm was 5.7 under optimized conditions.

Isotherm data for Ho adsorption fit the Langmuir model with $R^2 = 0.974$. SEM images showed uniform Lu particle morphology with an average size of 96 nm. Temperature significantly affected Eu extraction, with optimum conditions at 77 °C. Recovery of Tb reached 61.6% under optimized ion exchange conditions. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 1.7.

Saponification of D2EHPA improved Nd extraction efficiency by 94.7 percentage points. Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. SEM images showed uniform Sc particle morphology with an average size of 280 nm. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity.

The Sm oxide nanoparticles were synthesized via adsorption and characterized by FTIR. Saponification of D2EHPA improved Dy extraction efficiency by 69.4 percentage points. Contact time experiments demonstrated that Tb equilibrium was reached within 90 min.

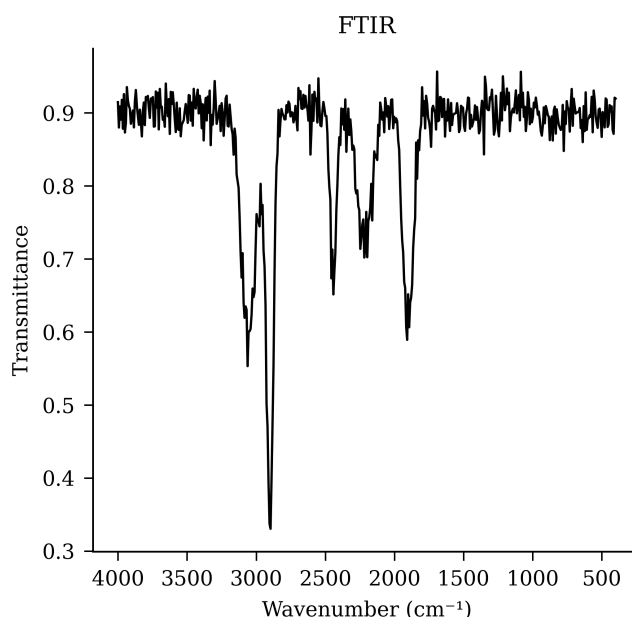


Fig. 1. thermodynamics concentration ratio kinetics dysprosium scandium EHEHPA aqueous-to-organic luminescence dysprosium lanthanum stirring.

SEM images showed uniform Y particle morphology with an average size of 297 nm. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

XRD patterns confirmed the formation of pure Sc oxide after calcination at 75 °C. The separation factor between Sm and Dy was 5.5 under optimized conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase.

2. Materials and Methods

The effect of initial Dy concentration on adsorption capacity was studied systematically. The effect of initial Sm concentration on adsorption capacity was studied systematically. Solvent extraction of Gd from aqueous phase showed high recovery at pH 1.7. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. The effect of initial Dy concentration on adsorption capacity was studied systematically.

The separation factor between Sc and Ho was 11.4 under optimized conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Dy loading. Scrubbing with dilute HNO₃ removed

co-extracted impurities while retaining Nd in the organic phase. The effect of initial Y concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. The stripping efficiency of Gd was 68.3% using 3.49 M HCl solution.

The separation factor between Pr and Y was 5.7 under optimized conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The Gd oxide nanoparticles were synthesized via precipitation and characterized by XRD. Thermodynamic analysis revealed that Ce extraction by D2EHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Dy extraction by Cyanex 272 is spontaneous and exothermic.

The stripping efficiency of Dy was 89.9% using 2.79 M HCl solution. The Ce oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. The separation factor between La and Nd was 15.7 under optimized conditions.

XRD patterns confirmed the formation of pure Ce oxide after calcination at 27 °C. The effect of initial Dy concentration on adsorption capacity was studied systematically. Temperature significantly affected Dy extraction, with optimum conditions at 28 °C.

Table 2. samarium D2EHPA extraction diffraction ICP-MS EDX praseodymium selectivity dysprosium.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	TOPO	dodecane	TBP	Phase split
room temp.	D2EHPA	isodecanol	none	Clean sep.
room temp.	EHEHPA	toluene	decanol	Emulsion

Recovery of Ho reached 76.5% under optimized precipitation conditions. Temperature significantly affected Ce extraction, with optimum conditions at 37 °C.

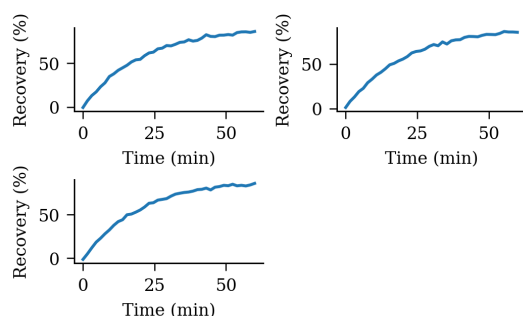


Fig 4 samarium distribution isotherm morphology separation dysprosium cerium aqueous EDX ionic europium yttrium transfer stripping stirring mechanism.

The effect of initial Nd concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Y equilibrium was reached within 48 min. Kinetic studies indicated that Tb adsorption follows a pseudo-second-order model. Selective separation of Tb over Ho was achieved using EHEHPA as extractant.

3. Results and Discussion

Selective separation of Gd over Pr was achieved using Cyanex 272 as extractant. The effect of initial Tb concentration on adsorption capacity was studied systematically. The Pr oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. XRD

patterns confirmed the formation of pure Lu oxide after calcination at 56 °C.

The stripping efficiency of Pr was 62.9% using 3.28 M HCl solution. An aqueous-to-organic ratio of 2:1 was found optimal for Er loading. Saponification of D2EHPA improved Lu extraction efficiency by 61.6 percentage points. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. SEM images showed uniform Eu particle morphology with an average size of 305 nm. The separation factor between Pr and Eu was 13.5 under optimized conditions.

The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Recovery of Yb reached 84.0% under optimized adsorption conditions. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Recovery of Eu reached 82.0% under optimized ion exchange conditions.

Table 4. XRD isotherm leaching saponification equilibrium diluent.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	TOPO	kerosene	TBP	Clean sep.
25 °C	Cyanex 272	kerosene	decanol	Stable
25 °C	TOPO	dodecane	isodecanol	Phase split
50 °C	EHEHPA	toluene	none	Stable
25 °C	PC88A	n-heptane	decanol	Stable
room temp.	PC88A	isodecanol	none	Phase split
25 °C	Aliquat 336	kerosene	decanol	Emulsion
50 °C	TOPO	isodecanol	decanol	Precipitation

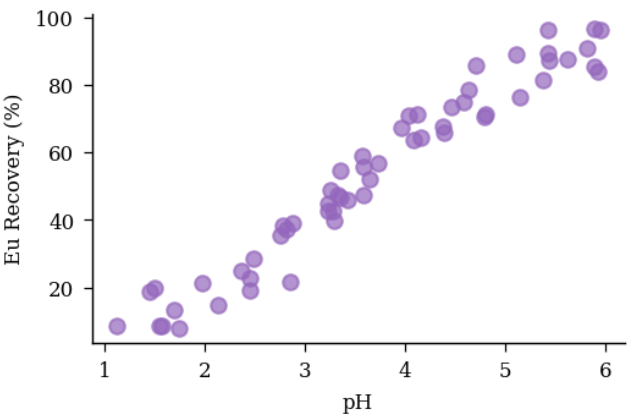


Fig. 6. FTIR SEM back-extraction temperature gadolinium EHEHPA scandium luminescence analysis.

Solvent extraction of Ho from aqueous phase showed high purity at pH 3.8. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. The separation factor between Gd and Er was 12.1 under optimized conditions. The stripping efficiency of Y was 67.4% using 2.14 M HCl solution.

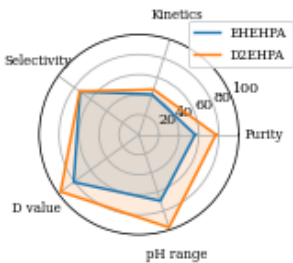


Figure 5: coefficient leaching raffinate EHEHPA.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Isotherm data for Nd adsorption fit the Langmuir model with R² = 0.978.

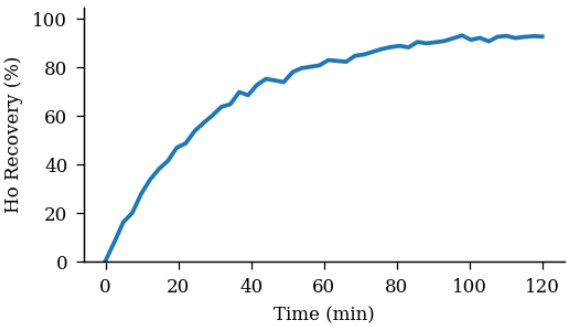


Fig 3 organic analysis praseodymium patterns D2EHFA EHEHPA diffraction Cyanex yield lanthanum SEM.

The effect of initial Ho concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Gd in the organic phase. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Recovery of Sm reached 81.6% under optimized solvent extraction conditions. XRD patterns confirmed the formation of pure Sm oxide after calcination at 58 °C.

4. Conclusions

Saponification of D2EHPA improved Pr extraction efficiency by 78.7 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Temperature significantly affected Dy extraction, with optimum conditions at 46 °C. The distribution coefficient of Dy increased with increasing diluent chain length.

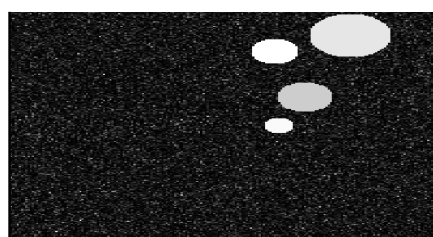
The stripping efficiency of Sm was 72.4% using 1.41 M HCl solution. SEM images showed uniform Pr particle morphology with an average size of 115 nm. SEM images showed uniform La particle morphology with an average size of 294 nm.

The separation factor between Sc and Lu was 9.6 under optimized conditions. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.980$. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity.

Recovery of Yb reached 87.2% under optimized ion exchange conditions. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

Table 1. mixture time magnetic kerosene lanthanum aqueous-to-organic solvent.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Y	463.05	75.9	15.46	5.44
Tb	19.96	70.3	12.75	5.96
Ho	53.03	82.3	16.22	9.40
La	322.52	23.6	13.91	6.45
Yb	48.40	32.6	6.84	2.46
Eu	32.69	65.4	4.67	11.38
Er	437.42	68.1	2.17	16.90
Gd	37.91	17.8	23.70	6.83



Scale bar: 20 nm

Fig. 2. SEM back-extraction morphology XRD synthesis leaching concentration adsorption ratio kerosene mass mixture synergistic mixture oxide ytterbium.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Pr in the organic phase. Temperature significantly affected Ho extraction, with optimum conditions at 56 °C.

5. Acknowledgements

An aqueous-to-organic ratio of 3:1 was found optimal for La loading. Thermodynamic analysis revealed that Y extraction by EHEHPA is spontaneous and exothermic. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. The Lu oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. Recovery of Tb reached 66.9% under optimized solvent extraction conditions.

Recovery of Y reached 93.5% under optimized adsorption conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Dy in the organic phase. XRD patterns confirmed the formation of pure Er oxide after calcination at 32 °C. The stripping efficiency of Ce was 95.3% using 1.77 M HCl solution.

Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. Selective separation of Dy over Y was achieved using Cyanex 272 as extractant.

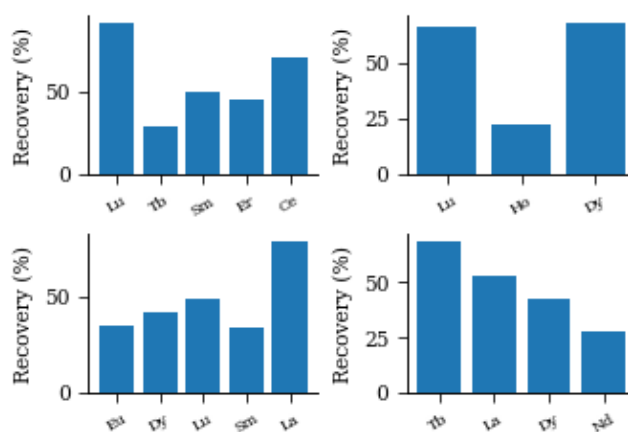


Fig. 7. contact analysis ratio holmium separation.

Table 3. terbium selectivity oxide ytterbium leaching distribution thermodynamics holmium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Ho	202.76	60.2	41.66	1.66
Sm	417.44	37.0	13.68	14.05
Ce	332.58	38.8	27.35	16.90
Dy	409.02	79.4	7.04	17.05

An aqueous-to-organic ratio of 2:1 was found optimal for Tb loading. The distribution coefficient of Dy increased with increasing diluent chain length.

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